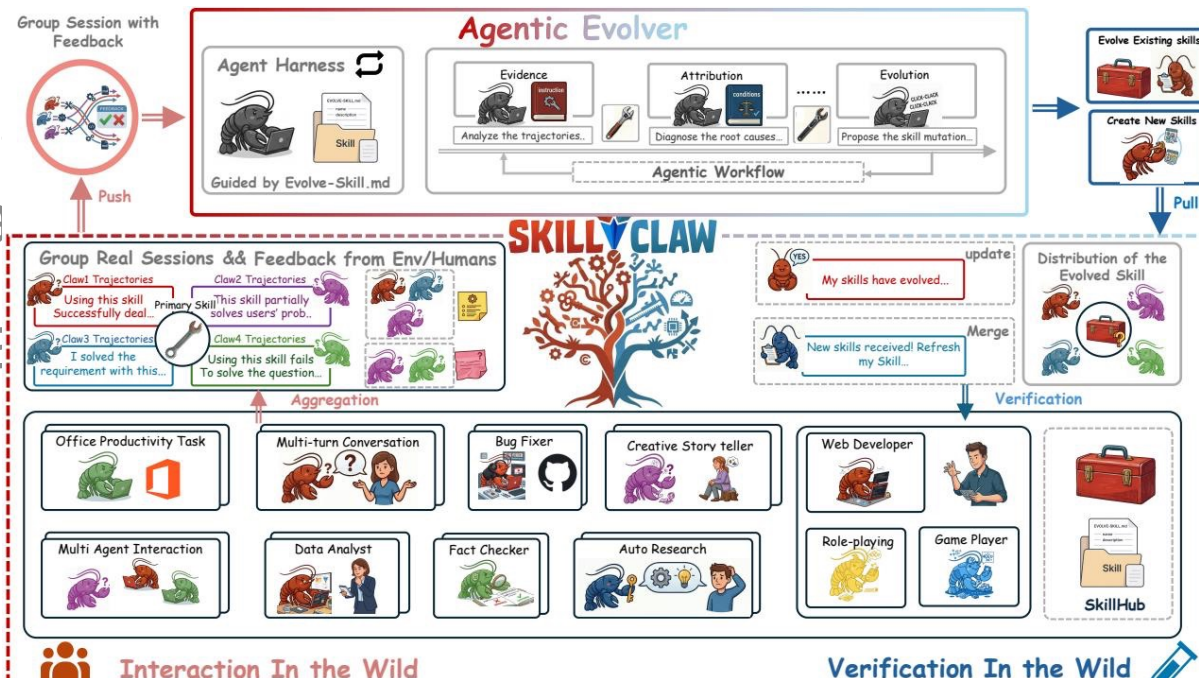


SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective

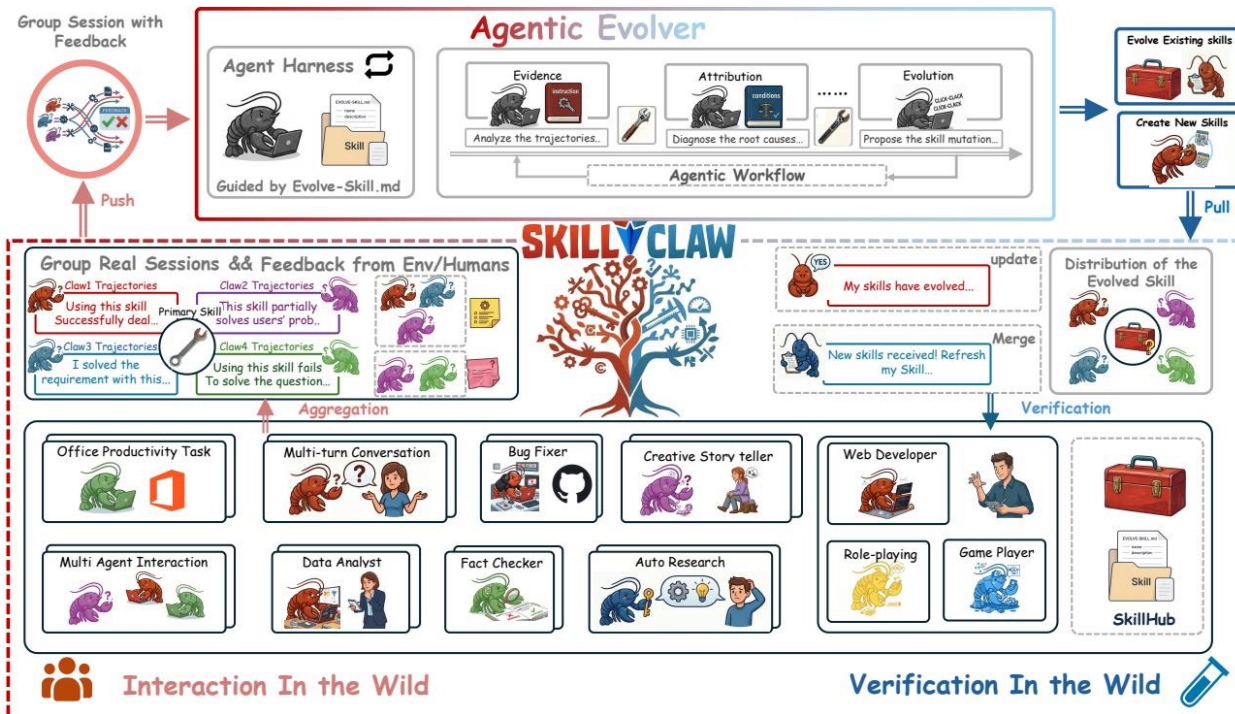
Source: a



Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

- Current LLM agent skills are manually installed and rarely updated from runtime experience.
 - Failures and solutions are rediscovered repeatedly across users, preventing system-level knowledge accumulation.
- Limitations of existing approaches:
 - Memory-based methods: store past trajectories but remain tied to specific instances.
 - Skill-based methods: compress experience into structured instructions but treat the resulting library as a static resource.

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Multi-user interaction → Session collection → Skill evolution → Skill synchronization
Four Key Stages:

- Evidence: Aggregating diverse trajectories
- Attribution: Diagnosing root causes
- Evolution: Agent-driven skill mutation
- Distribution: Verifying and syncing shared skills

Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

- The agentic evolver analyzes each skill-specific group through three stages:
 - 1. Evidence: Analyze interaction trajectories to identify behavioral patterns across users.
 - 2. Attribution: Diagnose root causes of failures and successes based on the gathered evidence.
 - 3. Evolution: Propose skill mutations by refining existing skills or creating new ones.
- **Key Paradigm Shift:**
 - Updates are produced through open-ended agent reasoning over evidence, rather than predefined

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

- Individual user interactions provide complementary behavioral signals.
 - Different users exercising the same skill under diverse contexts produce a unified view of that skill's behavioral boundary.
 - Reveals both the conditions under which it works and those under which it breaks.
- A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.
- Compatible with a wide range of Claw-style

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

Day	Candidate Skill	Skill Function	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	Reject	Not admitted to next cycle

Highlight: Only 1 of 6 candidates was accepted, illustrating the conservative verification process that prevents regressions.

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	Accept	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	Accept	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	Reject	Not admitted to next cycle

Highlight: 3 of 6 days yielded accepted updates, showing successful iterative refinement before reaching a plateau.

Key Takeaways: Three Properties That Distinguish SkillClaw from Existing Systems

- **1. Collective Evolution:**
 - Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.
- **2. Fully Automatic:**
 - Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.
- **3. Agentic Evolution Paradigm:**
 - Skill updates are produced through open-ended reasoning over evidence rather than predefined